**Table 3.2 Döbereiner Triads**

| S. No. | Elements in the Triad | Atomic weight of middle element | Average atomic weight of the remaining elements |
|---|---|---|---|
| 1 | Li, Na, K | 23 | $\frac{7+39}{2} = 23$ |
| 2 | Cl, Br, I | 80 | $\frac{35.5+127}{2} = 81.25$ |
| 3 | Ca, Sr, Ba | 88 | $\frac{40+137}{2} = 88.5$ |

This concept can not be extended to some triads which have nearly same atomic masses such as [Fe, Co, Ni], [Ru, Rh, Pd] and [Os, Ir, Pt].

In 1862, A. E. B. de Chancourtois reported a correlation between the properties of the elements and their atomic weights. He said 'the properties of bodies are the properties of numbers'. He intended the term numbers to mean the value of atomic weights. He designed a helix by tracing at an angle 45° to the vertical axis of a cylinder with circumference of 16 units. He arranged the elements in the increasing atomic weights along the helix on the surface of this cylinder. One complete turn of a helix corresponds to an atomic weight increase of 16. Elements which lie on the 16 equidistant vertical lines drawn on the surface of cylinder shows similar properties. This was the first reasonable attempt towards the creation of periodic table. However, it did not attract much attention.

In 1864, J. Newland made an attempt to classify the elements and proposed the law of octaves. On arranging the elements in the increasing order of atomic weights, he observed that the properties of every eighth element are similar to the properties of the first element. This law holds good for lighter elements up to calcium.

**Table 3.3 Newlands' Octaves**

| | | | | | | |
|---|---|---|---|---|---|---|
| $^{7}Li$ | $^{9}Be$ | $^{11}B$ | $^{12}C$ | $^{14}N$ | $^{16}O$ | $^{19}F$ |
| $^{23}Na$ | $^{24}Mg$ | $^{27}Al$ | $^{29}Si$ | $^{31}P$ | $^{32}S$ | $^{35.5}Cl$ |
| $^{39}K$ | $^{40}Ca$ | | | | | |

### 3.1.1 Mendeleev's Classification

In 1868, Lothar Meyer had developed a table of the elements that closely resembles the modern periodic table. He plotted the physical properties such as atomic volume, melting point and boiling point against atomic weight and observed a periodical pattern.

During same period Dmitri Mendeleev independently proposed that "the properties of the elements are the periodic functions of their atomic weights" and this is called periodic law. Mendeleev listed 70 elements, which were known till histime in several vertical columns in order of increasing atomic weights. Thus, Mendeleev constructed the first periodic table based on the periodic law.



11th Std Chemistry 068-099.indd 70 23-12-2021 15:10:10